

Desk Number _____

Student Number _____

Student Name _____



School of Science and Engineering

FINAL EXAMINATION (B)

Semester 2, May, 2021

CSC1001 Introduction to Computer Science

Examination Duration: 120 minutes

Reading Time: 10 minutes

This examination has 3 questions.

Exam Conditions:

This is a FORMAL Examination

This is a RESTRICTED OPEN BOOK Exam. Maximum of one (1) sheet of handwritten notes double sided are permitted

Materials Permitted In The Exam Venue:

Maximum of one (1) sheet of handwritten notes double sided are permitted. **NO OTHER MATERIALS PERMITTED**

Any calculators without the functionalities of programming and file storage are permitted.

Materials To Be Supplied To Students:

1 × 9 Page Answer Booklet

Question 1. (10 × 3% = 30%)

Pick the correct option in each of the following sub-questions. Note that only ONE option is correct.

1.1) Binary number ^{64 32 8 4 2 1}1101111.111 and hexadecimal number 5FC.3 equal to decimal numbers: **A**

- ☒ A. 111.875 and 1532.1875
☐ B. 110.85 and 1532.1875
☐ C. 111.111 and 1712.1875
☐ D. 110.85 and 1712.375

0.5 + 0.25 + 0.125

20.75 + 0.125

20.875

1.2) Concerning Python language, which of the following statements is incorrect? **C**

- A. Python uses an interpreter to translate source codes into machine codes. ✓
 B. In Python, reserved words cannot be used as variable names. ✓
☒ C. In Python, a variable has a fixed location in the memory. ✗
 D. **true** and **false** are not reserved words in Python. ✓
 mainly True, False

1.3) Concerning the following program, which of the following statements is incorrect? **C**

```
if x < 2:
    print('Below 2')
elif x < 20:
    print('Below 20')
elif x < 10:
    print('Below 10')
else:
    print('Something else')
print('Finished')
```

- A. **print('Below 2')** will be executed when $x = 1$.
 B. **print('Below 20')** will be executed when $x = 10$.
☒ C. **print('Below 10')** will be executed when $x = 8$. ✗
 D. **print('Something else')** will be executed when $x = 20$.

1.4) Concerning the object in Python, which of the following statements is incorrect? **D**

- ☒ A. Every object in Python has a unique ID. ✓
☐ B. The type of an object is determined automatically by Python interpreter according to its value. ✓
☐ C. Every variable is essentially a reference to an object. ✓
☐ D. The ID of an object may be changed during the execution of the program. ✗

1.5) Concerning stack and queue, which of the following statement is correct? **A**

- ☒ A. A stack can be accessed based on the “last in first out” principle and the time complexity of inserting an element into a stack is constant. ✓

- B. More than one elements of a stack can be accessed simultaneously if necessary. α
- C. Elements can be inserted at any time in a queue, but only the element that has been in the queue the shortest can be next removed. α
- D. The time complexity of removing an element from a stack or a queue is linear α

const
(O(1))

1.6) Concerning the following program, which of the following statements is incorrect?

C

```
class A:
    def __init__(self, a, b):
        self.a = a
        self.b = b

    def multiply(self):
        x = a*b
        return x

    def add(self):
        x = a+b
        return x
```

- A. Class A() contains two newly defined data fields. $\checkmark$
- B. The scope of **self** is throughout the whole body of class A(). $\checkmark$
- C. Method **add()** will return the sum of two data fields **a** and **b**. α on stop self.a+self.b (error inside a, b init loc var)
- D. Variable **x** defined in **multiply()** and **add()** are two different local variables. $\checkmark$

1.7) Concerning the following two programs, which of the following statements is incorrect? B

```
class A:
    def A(self):
        radius = 3
        print(radius)

def main():
    a = A()
    a.A()
```

```
class A:
    def __init__(self):
        radius = 3

    def setRadius(radius):
        self.radius = radius

def main():
    a = A()
    a.setRadius(3)
```

- A. When executing the first program, an error will be triggered. α $\checkmark$ No initializer (self)
- B. When executing the second program, an error will be triggered. $\checkmark$
- C. The first class A() will inherit the initializer from its parent class. α
- D. The **a.setRadius(3)** statement of the second program aims to set the radius of **a** to 3. $\checkmark$

1.8) Concerning private members of a class, which of the following statements is incorrect? C

- A. In Python, the names of private data fields and private methods start with 2 underscores and do not end with 2 underscores. $\checkmark$
- B. Private data fields can be accessed within a class. $\checkmark$
- C. Private methods can be directly accessed outside the class definition. α
- D. Private data fields are defined to prevent potential data corruption. $\checkmark$

1.9) Concerning the following program, which of the following statements is incorrect?

A

```
class Student:
    def __str__(self):
        return 'Student'

    def printStudent(self):
        print(self.__str__())

class GraduateStudent(Student):
    def __str__(self):
        return 'Graduate student'

def main():
    g = GraduateStudent()
    print(g.__str__())

main()
```

- ~str~ Overriden in subclass*
- A. The output of this program is 'Student'. ☒
 - B. Class `GraduateStudent()` has inherited the method `printStudent()`. ☒
 - C. Method `__str__()` has been overridden in class `GraduateStudent()`. ☒
 - D. In class `GraduateStudent()`, method `printStudent()` will be inherited from class `Student()`. ☒

1.10) Concerning the binary search algorithm, which of the following statement is correct? A

A

- A. This algorithm is able to find out whether a target element exists in a given sequence of elements. ☒
- B. Binary search algorithm can be applied on an unsorted sequence. ☒
- C. The time complexity of binary search is $O(n)$. ☒ *$O(\log n)$*
- D. Binary search is usually equally efficient compared with sequential search. ☒ *better*

Question 2. ($10 \times 4\% = 40\%$)

Pick the correct option/s in each of the following sub-questions. Note that there may be MULTIPLE correct options for each sub-question.

2.1) Concerning recursion, which of the following statement/s is/are correct? BCD

BCD

- A. When a function makes two or more recursive calls, this function ~~does not~~ belong to multiple recursion. ☒
- B. A recursive algorithm should have one or more base cases which is or are non-recursive. ☒
- C. To write a recursion program, we usually need to define the problem in a recursive way first. ☒
- D. The base case in recursive algorithm can return nothing. ☒

- 2.2) Concerning the following program, which of the following statement/s is/are correct? A

A

```

a=1
b=2
class A:
    def __init__(self):
        self.a = 10
        self.b = 20
    def B(self):
        a = 100
        print(a)
def main():
    obj = A()
    print(obj.a)
    obj.B()
    print(a)
main()

```

- A. In this program, three different variables **a** have been defined. ✓
 B. Data field **a** of class **A()** can only be accessed within the class definition. ✗ obj.a 98%
 C. The value of global variable **a** will initially be set as 1, and then changed to 100. ✗
 D. The output of this program is

10
 100
~~100~~ 1

9

- 2.3) Given the program, which of the following statement/s is/are correct? AC

AC

```

def function(input):
    input = input * 2
    return input

output1 = function('32')
output2 = function(32)
output3 = function(32.0)

```

- A. The data types of output1, output2, output3 are: str, int, float. ✓
 B. output1 is exactly: 64 ✗ 3232
 C. output2 is exactly: 64 ✓
 D. output3 is exactly: 64 ✗ 64.0

- 2.4) Concerning algorithm analysis, which of the following statement/s is/are correct?

ABC

AB

- A. The big-Oh notation allows us to say that a function $f(n)$ is always less than or equal to another function $g(n)$ up to a constant factor when n is large enough. ✓
- B. The big-Oh notation is usually used to characterize the running time of an algorithm in the asymptotic sense. ✓
- C. Function $5n^4 + 6n^3 + 2n^2 + 2n + 2n$ is $O(n^4)$. ✓
- D. When we analyse an algorithm, we are usually interested at its average performance regardless of the input size. ✗

2.5) Assume that the five numbers 1, 2, 3, 4, and 5 are pushed onto the stack, which of the following statement/s is/are correct?: CD

- A. If the numbers are pushed in the order of 1-2-3-4-5, then the first number popped will be $\times 5$ ✗
- B. In Python, these five numbers are essentially saved in a list. *Nein bei Array/linked list* ✗
- C. The stack can be implemented based on a dictionary. ✓
- D. The time complexity of popping out all elements in the stack is $O(n)$. ✗

2.6) Concerning the binary search algorithm, which of the following statement/s is/are correct? ACD

- A. The purpose of this algorithm is to find out whether a target element exists in a given sequence of elements. ✓
- B. Binary search algorithm can be applied on an unsorted sequence. ✗
- C. The time complexity of binary search is $O(\log n)$. ✓
- D. Binary search is usually more efficient than sequential search. ✓

2.7) Concerning stack, which of the following statement/s is/are correct? AD

- A. A stack can be accessed based on the "last in first out" principle. ✓
- B. More than one elements of a stack can be accessed simultaneously if necessary. *One at a time* ✗
- C. The time complexity of Inserting an element into a stack is *linear*. ✗
- D. The time complexity of removing an element from a stack is constant. ✓

2.8) Concerning queue, which of the following statement/s is/are correct? A

- A. Data are saved sequentially in a queue. ✓
- B. Data can only be removed from the *head* tail of a queue. ✗
- C. Data can only be inserted at the *tail* head of a queue. ✗
- D. A queue can only be implemented based on a *list*. ✗

2.9) Concerning linked list, which of the following statement/s is/are correct? ABD

- A. We can identify the tail of a singly linked list as the node having **None** as its next reference. ✓
- B. The time complexity of inserting a node at the tail of a singly linked list is constant. ✓
- C. The time complexity of removing a node at the tail of a singly linked list is constant. ✗
- D. Every node in a doubly linked list contains at least two references. ✗

2.10) Concerning a binary tree, which of the following statement/s is/are correct? ABC

- A. In a binary tree, data are saved hierarchically. ✓
- B. In a non-empty binary tree, there is only one node which has no parent. *(root)* ✓
- C. Every node on a binary tree contains at least three references. *parent / child nodes* ✗

D. In a binary tree, every node has zero or two child nodes. ~~X~~
only left or right

Question 3. (10% + 8% + 12% = 30%)

Answer the following questions.

3.1) Concerning the following program

```
def bubble(bubbleList):
    listLength = len(bubbleList)
    while listLength > 0:
        for i in range(listLength - 1):
            if bubbleList[i] < bubbleList[i+1]:
                buf = bubbleList[i]
                bubbleList[i] = bubbleList[i+1]
                bubbleList[i+1] = buf
            listLength -= 1
        return bubbleList

def main():
    bubbleList = [3, 4, 1, 2, 5, 6, 0, 10, 7]
    print(bubble(bubbleList))
main()
```

(n-1) + (n-2) + ... + 1

*นี่คือการสลับ
→ เรียงจากน้อยไปมาก*

Answer the following questions:

- What are the outputs of this program?
- Which sorting algorithm is applied in this program?
- What is the time complexity of this sorting algorithm? $\rightarrow O(n^2)$

Answer:

- [10,7,6,5,4,3,2,1,0] 3 points
- bubble sort 3 points
- $O(n^2)$ 4 points

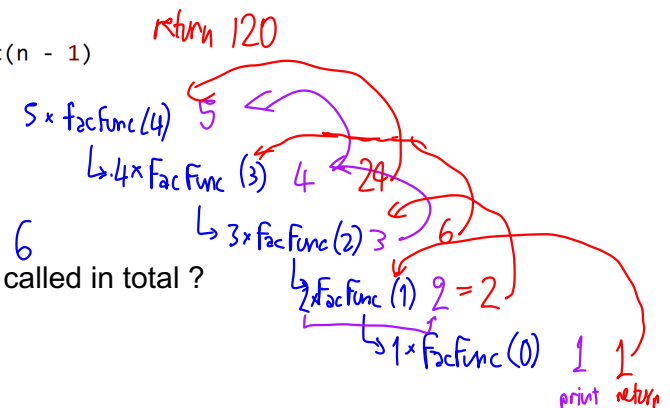
3.2) Concerning the following program

```
def facFunc(n):
    if n < 0:
        print('Invalid input.')
        return None
    elif n == 0:
        return 1
    else:
        a = n * facFunc(n - 1)
        print(n)
        return a
print(facFunc(5))
```

Answer the following questions:

- How many times function `facFunc()` will be called in total ?
- What are the outputs of this program?

Answer:



1) 6 3 points

2) 5 points

1

2

3

4

5

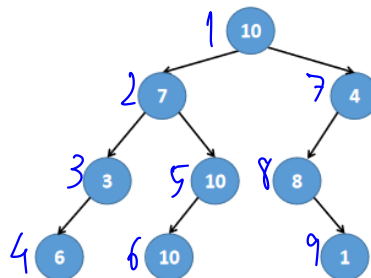
120

3.3) Concerning the following program, and assume that input **t** will be a reference pointing to the root of a binary tree.

```
def DFSearch(t):
    if t:
        print(t.element)
        if (t.left is None) and (t.right is None):
            return
        else:
            if t.left is not None:
                DFSearch(t.left)
            if t.right is not None:
                DFSearch(t.right)
```

Answer the following questions:

- Which algorithm is implemented in this function?
- What is the time complexity of this algorithm? *ความซับซ้อน → O(n)*
- Is this function a linear recursive or multiple recursive function?
- If input **t** is referencing to the root of the following tree, what would be the outputs of this function?



Answer:

1) Depth First Search 3 points

2) O(n) 3 points

3) multiple recursive function 3 points

recur 2 sou

4) 10

3 points

7

3

6

10

10

4

8

1

END OF EXAMINATION